

Kapture Finance

Outbound Voice AI Collections Agent — "Maya"

High-Level Design Document

Version 2.0 (two-step token-gated auth) • August 2026

Prepared for: AI Delivery Take-Home Assignment

0. Scope & Objective

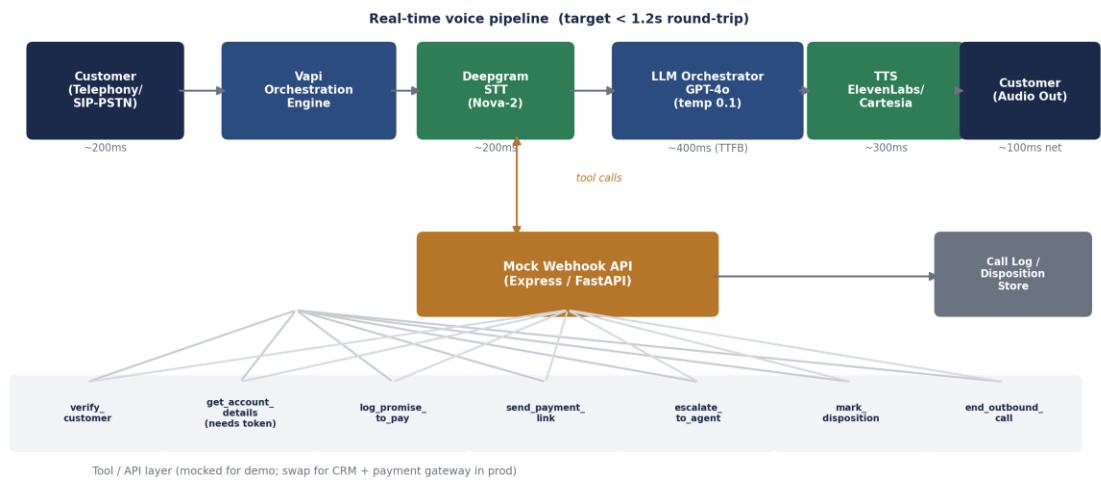
This document is the engineer-ready design for "Maya", an outbound voice AI collections agent for Kapture Finance. Maya calls customers with an overdue loan EMI, authenticates them via a two-step, token-gated process, discloses the overdue amount only after both authentication steps succeed, negotiates a resolution, and logs a disposition on every call — including calls that never reach authentication — without a human agent on routine calls.

Reference scenario used throughout this document:

- Customer: Rahul Sharma • Personal Loan • Account ACC-88392
- Overdue amount: ₹8,499 • 12 days past due (DPD)
- Channel: outbound voice call via Vapi, built on Deepgram STT, GPT-4o orchestration, and ElevenLabs/Cartesia TTS

1. Architecture & Pipeline

The call flows through a standard real-time voice pipeline: Telephony (SIP/PSTN) → STT → LLM Orchestrator → TTS → Telephony output, with a mocked webhook API acting as the tool/action layer and a call-log datastore for dispositions.



1.1 Latency Budget (target: < 1.2s end-to-end round-trip)

Hop	Component	Budget	Notes
1	STT (speech recognized)	~200 ms	Deepgram Nova-2, streaming/interim results enabled
2	LLM first byte (TTFB)	~400 ms	GPT-4o, temperature 0.1, short system + few-shot state context
3	TTS synthesis start	~300 ms	ElevenLabs/Cartesia streaming synthesis, not full-utterance wait
4	Network / telephony overhead	~200 ms	SIP trunk + Vapi orchestration overhead, round trip
Total		~1.1 s	Within the 1.2s target; tool calls (verify_customer etc.) add 150–400ms on top only on turns that need them

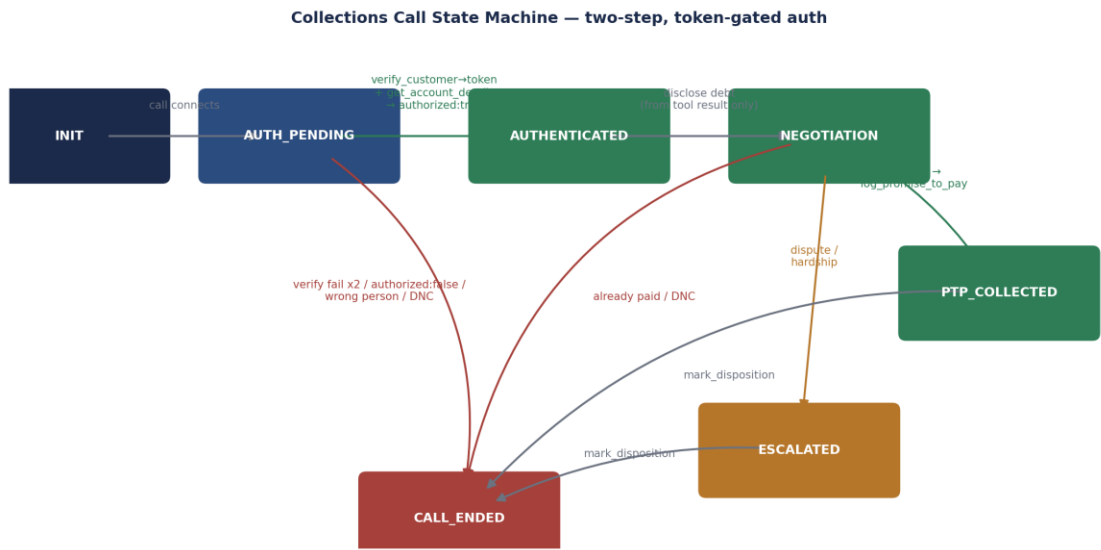
Mitigations if budget is exceeded: enable STT interim/partial results to start LLM reasoning earlier, keep the system prompt lean per state instead of one giant prompt, cap tool-call payloads, and use streaming TTS so audio starts before the full response is generated.

2. Conversation Flow / State Machine

Authentication is a two-step, token-gated process, and it is state-enforced rather than prompt-discretionary: the orchestrator layer blocks any tool or response branch that would disclose debt information unless BOTH steps have succeeded in the current call.

- Step 1 — `verify_customer(account_id, verification_code)` must return `verified:true`. On success it issues a short-lived, single-call `auth_token`.
- Step 2 — `get_account_details(account_id, auth_token)` must return `authorized:true`, using the exact token from Step 1. This is the only call that returns actual debt figures.

A tool call "executing successfully" (HTTP 200, no error) is not the same as the business operation succeeding — the agent is explicitly instructed to inspect the `verified` / `authorized` / `success` field in every tool result before speaking about the outcome, never to assume success, and never to accept, invent, or repeat an `auth_token`.



2.1 States

State	Entry condition	Locked behavior
INIT	Call connects	No tools available yet; greeting only
AUTH_PENDING	Greeting complete, target customer confirmed	Only <code>verify_customer</code> is callable; debt words are hard-blocked; max 2 verification attempts
AUTHENTICATED	<code>verify_customer</code> → <code>verified:true</code> AND <code>get_account_details</code> → <code>authorized:true</code>	Unlocks debt disclosure using only fields returned by <code>get_account_details</code> ; max 1 entry per call
NEGOTIATION	Debt disclosed	Intent classification branches to PTP / already-paid / hardship / dispute / DNC / callback / hostile
PTP_COLLECTED	<code>log_promise_to_pay</code> returns <code>success:true</code>	<code>send_payment_link</code> unlocks only if the customer asks for a link; no duplicate PTP calls for the same commitment
ESCALATED	Dispute, hardship, or explicit human request	<code>escalate_to_agent</code> required before <code>CALL_ENDED</code>

State	Entry condition	Locked behavior
CALL_ENDED	Any terminal path, with no exceptions	mark_disposition must have fired exactly once — including calls that never reach AUTHENTICATED

Hard rule: AUTH_PENDING → AUTHENTICATED is strictly locked behind two successful, in-call tool results — verify_customer(verified:true) followed by get_account_details(authorized:true) using that exact token. The LLM cannot self-report a customer as verified or authorized without both tool results, and any pre-tool-call debt disclosure — or a call that ends without a mark_disposition — is treated as a policy violation to flag in observability.

3. Intents & Entities

Intent	Description	Typical state
Confirm_Identity	Caller confirms/denies being the target customer	INIT / AUTH_PENDING
Provide_Verification	Caller supplies PAN-4 / birth year	AUTH_PENDING
Promise_To_Pay	Caller commits to a payment date/amount	NEGOTIATION
Already_Paid	Caller claims the debt is already settled	NEGOTIATION
Hardship_Claim	Caller cites inability to pay (job loss, medical, etc.)	NEGOTIATION
Dispute_Debt	Caller disputes the amount or existence of the debt	NEGOTIATION
Request_DNC	Caller asks to be removed from calling lists	any state
Wrong_Person	Caller is not the target customer	INIT / AUTH_PENDING
Request_Human	Caller explicitly asks for a human agent	any state

3.1 Entities extracted

Entity	Type	Notes
PTP_Date	ISO-8601 date	Normalized from natural language ("this Friday", "parso") to a concrete date
PTP_Amount	Number	Defaults to full overdue amount unless a partial amount is explicitly agreed and within the 10% waiver cap
Verification_Code	String (4 digits)	Last 4 digits of PAN or 4-digit birth year
Hardship_Reason	String	Free text passed to escalate_to_agent, not stored with raw PII beyond what's needed
Language	Enum (en, hi, hi-en)	Drives mid-call TTS/STT language switch

4. Tool / API Specifications

Full JSON Schemas are in vapi/tool_definitions.json. Summary below — note that every tool after get_account_details requires the exact auth_token issued by verify_customer in that same call:

Tool	Purpose	Key inputs	Key outputs
verify_customer	Step 1 of auth: identity check	account_id, verification_code	verified (bool), auth_token, message

Tool	Purpose	Key inputs	Key outputs
get_account_details	Step 2 of auth: gates all debt disclosure	account_id, auth_token	authorized (bool), overdue_amount, days_past_due, loan_type
log_promise_to_pay	Records PTP commitment	account_id, auth_token, ptp_date, amount	success, ptp_id
send_payment_link	Sends SMS/WhatsApp payment link, only on request	account_id, auth_token, channel	success, message
escalate_to_agent	Hands off to a human	account_id, auth_token (if authenticated), reason, notes	success, ticket_id
mark_disposition	Final call outcome; called exactly once, always	account_id, auth_token (if authenticated), status, notes	success, timestamp
end_outbound_call	Ends the call; only after mark_disposition succeeds	—	success

Disposition status enum: PTP_AGREED, ALREADY_PAID, DISPUTED, HARDSHIP_ESCALATED, WRONG_PERSON, DO_NOT_CALL, VERIFICATION_FAILED, CALLBACK_REQUEST, HOSTILE, NO_RESPONSE. Escalation reason enum: DISPUTE, HARDSHIP_REQUEST, CUSTOMER_REQUEST, LANGUAGE_UNSUPPORTED.

5. Auth & Data Safety Protocols

- Two-step, token-gated disclosure: no debt, loan, or EMI word is spoken until BOTH verify_customer returns verified:true AND get_account_details returns authorized:true using that exact token, in the current call — enforced at the orchestrator/state layer, with the LLM prompt as a second line of defense, not the only one.
- Tokens are short-lived (10-minute expiry in the mock server), single-call, and single-use — issued only by verify_customer, never accepted from the customer, never invented by the agent, and never spoken aloud. A token is invalidated once mark_disposition fires for that call.
- Verification attempts are capped at 2; a third guess is never accepted, nothing is disclosed, and the call moves straight to a graceful close with a VERIFICATION_FAILED disposition.
- Tool-result inspection is mandatory: "tool executed successfully" is never treated as equivalent to verified / authorized / success being true — the agent must read the actual field before claiming an outcome to the customer.
- PII masking in logs: customer names are masked (e.g. "Rahul S****") and auth tokens are truncated in all application logs; verification codes are never logged in plaintext.
- Least-privilege tool access: log_promise_to_pay, send_payment_link, and disclosure of any debt detail are only reachable after AUTHENTICATED state is reached (enforced by the orchestrator's allowed-tools-per-state list, not just prompt instruction).
- Data retention: call transcripts and dispositions are retained per Kapture Finance's data retention policy; recordings should be encrypted at rest with access limited to compliance/QA roles.

6. Guardrails & Compliance

6.1 Fair collections norms (RBI Fair Practices Code-aligned)

- Calling window enforced at the dialer/campaign level: 08:00–19:00 local time only.
- Mandatory self and company disclosure at call start ("This is Maya calling from Kapture Finance").

- No threats, harassment, or repeated pressure; a single polite follow-up per objection.
- Instant opt-out / Do-Not-Call handling — logged and actioned within the same call, not deferred.

6.2 Hallucination & policy guardrails

- The agent cannot offer a waiver greater than 10% of the outstanding amount under any circumstance.
- The agent cannot invent a tool result — it must wait for and use the actual webhook response, never assume success.
- The agent cannot make legal threats or reference credit bureau action as a threat.
- Off-topic requests (unrelated to this loan/call) are politely declined and redirected back to the call purpose.

7. Edge Cases Matrix

Case	Trigger	Agent behavior
Hostile / abusive user	Slurs, threats, shouting	One calm warning → if it continues, mark_disposition(HOSTILE) and end politely
Silent user / voicemail	No speech detected	Up to 2 re-prompts → mark_disposition(NO_RESPONSE); voicemail beep detected → short compliant message, no debt details, then hangup
Mid-call language switch	Caller switches English ⇌ Hindi	Agent mirrors the language while preserving state and disclosure rules; escalates if unsupported language is used
Wrong number / third party	Confirmed not the target customer	No debt disclosure; ask about reachability only; mark_disposition(WRONG_PERSON)
Failed verify_customer	2 incorrect verification attempts	No disclosure; mark_disposition(VERIFICATION_FAILED); end call
get_account_details unauthorized	Token missing/invalid/expired despite verified:true	No disclosure, no invented retry token; mark_disposition(VERIFICATION_FAILED); end call
Dispute	Caller denies the debt exists / amount is wrong	No argument; mark_disposition(DISPUTED); escalate_to_agent(DISPUTE)
Already paid	Caller claims prior payment	Capture date/mode; mark_disposition(ALREADY_PAID); explain 24–48h processing
Financial hardship	Caller cannot pay in full	Empathize, no unauthorized waivers/settlements; mark_disposition(HARDSHIP_ESCALATED); escalate_to_agent(HARDSHIP_REQUEST)
Callback / busy	Caller can't talk now or requests a callback	Ask preferred time, no promised slot unless tool-confirmed; mark_disposition(CALLBACK_REQUEST)
Do-Not-Call request	At any point in the call	Immediate acknowledgement, no auth requested, mark_disposition(DO_NOT_CALL), end call
Tool returns AUTH_REQUIRED / success:false	Any protected tool call	Never claim success; retry at most once for transient failures; offer human escalation if it still fails

8. Escalation & Disposition

Escalation to a human agent (escalate_to_agent) occurs for: disputes, hardship requests beyond standard partial-payment/extension options, explicit requests for a human, and unsupported languages. Hostile callers are de-escalated once and then ended, not escalated, unless they explicitly ask for a human.

Every call — regardless of path, including calls that never reach AUTHENTICATED — ends with exactly one mark_disposition call. Valid terminal statuses: PTP_AGREED, ALREADY_PAID, DISPUTED, HARDSHIP_ESCALATED, WRONG_PERSON, DO_NOT_CALL, VERIFICATION_FAILED, CALLBACK_REQUEST, HOSTILE, NO_RESPONSE. This single, mandatory write — enforced even on the failed-verification path — is what makes every call auditable end-to-end.

9. Observability Metrics

Metric	Definition	Why it matters
Containment Rate	% of calls resolved without human escalation	Core efficiency metric for the bot vs. human agents
PTP Rate	% of calls ending in a valid Promise-to-Pay	Primary business outcome metric
First Call Resolution (FCR)	% of calls with a clean, valid disposition logged	Signals conversation and tool-calling reliability
Auth Success Rate	% of AUTH_PENDING entries reaching AUTHENTICATED	Flags verification UX or fraud/testing issues
Pre-Auth Disclosure Violations	Count of any debt word spoken before verified:true	Hard compliance metric — target: zero, always
Avg. End-to-End Latency	Mean STT→LLM→TTS round trip per turn	Tracks against the 1.2s budget
Drop / Hangup Rate	% of calls ending abnormally (no disposition)	Signals bugs, timeouts, or bad UX

Appendix A — Sequence Diagram Source

Mermaid.js source for the sequence diagram (also included as docs/architecture_sequence.mmd in the repo):

```
sequenceDiagram
    autonumber
    actor Customer
    participant Tel as Telephony / SIP-PSTN
    participant Vapi as Vapi Orchestration Engine
    participant STT as Deepgram STT (Nova-2)
    participant LLM as LLM Orchestrator (GPT-4o)
    participant Srv as Mock Webhook API
    participant TTS as TTS (ElevenLabs/Cartesia)

    Customer->>Tel: Answers outbound call
    Tel->>Vapi: Stream audio
    Vapi->>STT: Real-time audio stream
    STT-->>Vapi: Transcribed text stream
```

```
rect rgb(240,240,240)

  note over Vapi,LLM: STATE 0-1 – Greeting & Auth Step 1 (no debt disclosed)

  Vapi->>LLM: Conversation state + transcript
  LLM-->>Vapi: "Confirm last 4 digits of PAN or DOB"
  Vapi->>TTS: Synthesize speech
  TTS-->>Customer: Play audio
  Customer->>Vapi: Speaks verification code
  Vapi->>LLM: Transcript
  LLM->>Srv: verify_customer(account_id, code)
  Srv-->>LLM: { verified: true, auth_token }

end
```

```
rect rgb(225,235,250)

  note over Vapi,LLM: Auth Step 2 – token-gated disclosure
  LLM->>Srv: get_account_details(account_id, auth_token)
  Srv-->>LLM: { authorized: true, overdue_amount, days_past_due }

end
```

```
rect rgb(220,245,220)

  note over Vapi,LLM: STATE 2-4 – Disclosure, Negotiation, Action
  LLM->>Vapi: Disclose debt (from tool result only) + ask intent
  Vapi->>TTS: Audio output
  TTS-->>Customer: Play audio
  Customer->>Vapi: States intent (e.g. "I'll pay Friday")
  LLM->>Srv: log_promise_to_pay(account_id, auth_token, ptp_date, amount)
  Srv-->>LLM: { success: true, ptp_id }
  LLM->>Srv: send_payment_link(account_id, auth_token, channel)
  Srv-->>LLM: { success: true }

end
```

```
rect rgb(250,235,215)

  note over Vapi,LLM: STATE 5 – Close & Disposition
  LLM->>Srv: mark_disposition(account_id, auth_token, status, notes)
  Srv-->>LLM: { success: true }
  LLM->>Srv: end_outbound_call()
  LLM->>Vapi: Polite goodbye

end
```


Vapi->>Customer: End call